

Building Footprint Inference for Building-to-Connection Mapping

Supporting Document — Prodegee x RMIT Data Science & AI Capstone 2026

About Prodegee

Prodegee is a purpose-driven technology company developing an AI-powered platform for climate finance in the Global South. Its mission: make investment-grade data accessible for the 760 million people who still lack electricity access, so that electrification planning and financing can be based on evidence rather than assumptions.

The Problem

Investment decisions in off-grid electrification are made using incomplete, inconsistent, and often outdated data. Survey-based demand estimation — the industry standard — overestimates actual electricity consumption by 3x to 3.75x (Blodgett et al., 2017). This leads to oversized infrastructure, failed financial projections, and increased risk perception among investors. The data gap is one of the primary barriers to scaling private capital into clean energy access: mini-grid developers struggle to secure financing because the data supporting their business cases is not credible to institutional investors.

What Prodegee Builds

Prodegee replaces survey-based estimates with geospatial intelligence. The platform processes satellite imagery, building footprints, facility datasets, and socioeconomic indicators through AI-powered pipelines to produce building-level electricity demand estimates for entire countries. These estimates flow through a sequence of modules:

1. **Geospatial data integration** — Merging building footprints, settlement data, and facility locations into a unified data layer, then determining which buildings share an electricity connection (the focus of these RMIT projects)
2. **Electrification planning (Ensembla)** — Costing all viable technology options (solar home systems, DC mesh-grids, mini-grids, grid extension) for every settlement cluster
3. **Financial projections** — Translating technical plans into investor-ready financial statements with ESG integration
4. **Institutional platform** — Connecting projects to institutional investors through a consent-based data sharing model that preserves project data sovereignty

Each module feeds the next — the quality of upstream geospatial data determines the reliability of everything downstream. The RMIT projects contribute to the first and most foundational layer.

Organisational Structure

Prodegee operates within a dual-entity architecture designed to protect its mission as it scales:

- **16plus1.org** — A planned Australian charity (Company Limited by Guarantee) that will serve as the constitutional guardian of the platform's 16 core ethical features covering data sovereignty, privacy, auditability, AI accountability, and transparency. These features will be legally embedded as constitutional articles that cannot be altered by commercial pressure. 16plus1.org has not yet been incorporated — it will be established as the platform approaches commercial launch.
- **Prodegee.com** — The commercial entity that builds and operates the platform, subject to 16plus1.org oversight once incorporated. The charity will retain majority control, ensuring that revenue generation remains aligned with the mission.

This structure means students are contributing to a project with genuine constitutional protections for its ethical commitments — not a startup that might pivot away from its stated mission.

Open Source Strategy

Prodegee follows a staged open-source strategy. All input data (satellite imagery, building footprints, facility datasets) is open. Individual algorithms and processing scripts are progressively open-sourced under a dual license (MIT/Apache 2.0 for community use, commercial license for enterprise deployments). The platform's workflow orchestration layer — how algorithms are sequenced and coordinated — remains proprietary until commercial sustainability is established. The direction is always toward openness, gated by commercial milestones rather than fixed timelines.

The Fiji Problem

In rural electrification planning, a foundational assumption — one building equals one electricity connection — causes systematic errors that cascade through every downstream calculation.

In practice, particularly in tropical and Pacific Island settings, but also across much of sub-Saharan Africa, multiple buildings typically belong to a single household or productive compound and share one electricity connection. A typical compound in West Africa might include a main dwelling, a kitchen building, a storage structure, and a latrine — all served by one meter. Over-counting connections leads to over-dimensioned distribution equipment, inflated cost estimates, and financially unsustainable projects.

We call this the **Fiji Problem** because it was in Fiji where we first encountered it while working with MIT-Comillas's Reference Electrification Model (REM). Neither REM nor KTH's OnSSET — the two dominant tools in geospatial electrification planning, both

developed by dedicated university research teams over multiple years — addresses building-to-connection mapping. Both assume a direct relationship between buildings (or population points) and electricity connections.

This is not a minor calibration issue. If a settlement has 200 buildings but only 60 actual electricity connections, over-counting by 3.3x cascades through every calculation: distribution network sizing, transformer capacity, generation requirements, battery storage, and financial projections. The resulting infrastructure is over-sized, over-priced, and often financially non-viable — reinforcing the perception that off-grid electrification is too risky for private investment.

Solving the Fiji Problem: Four Signals, Four Teams

Solving the Fiji Problem requires combining multiple evidence signals, each indicating whether buildings likely share a connection. Different signals require different analytical approaches:

Signal	Team	What It Tells Us
Temporal co-construction	DS-1	Buildings constructed at the same time likely belong to the same compound
Spatial clustering	DS-2	Natural clusters in building arrangements suggest shared compounds
Facility compound membership	DS-3	Known facility locations reveal institutional compounds
Building footprint characteristics	DS-4 (this team)	Building shape/size indicates use type and ancillary structures

Each team independently develops one signal through a two-stage methodology, then all four signals are combined in a collaborative integration workshop.

The Human-to-AI-to-Scale Methodology

Each Data Science & AI team follows the same methodological arc:

Stage 1 — Human Pattern Identification: Students analyse examples where ground truth is known, identify discriminating patterns and features, develop rule-based or statistical algorithms, and validate on held-out test data. Stage 1 produces accurate results for hundreds or thousands of examples.

Stage 2 — AI Model Training: Students use Stage 1 outputs as training labels to train machine learning models that recognise the patterns identified in Stage 1. The AI model can then infer patterns at scale — processing millions of buildings across entire countries, far beyond what humans can manually analyse.

This is the key insight: humans cannot manually analyse 10+ million buildings per country, but they can identify the patterns that AI learns to recognise. The AI extends human judgment, not replaces it.

This Team's Signal: Building Footprint Inference

DS-4 focuses on what buildings themselves reveal through their physical characteristics — area, aspect ratio, shape complexity, number of vertices, proximity to roads, and relative size within clusters.

A large rectangular building is likely institutional; a small square structure near a larger building is likely an ancillary outbuilding (kitchen, storage, latrine) that should not be counted as a separate electricity connection. The team develops classification models that distinguish primary residential buildings, ancillary outbuildings, commercial structures, institutional facilities, and agricultural buildings — then scales this classification to millions of buildings per country using AI.

Key research questions:

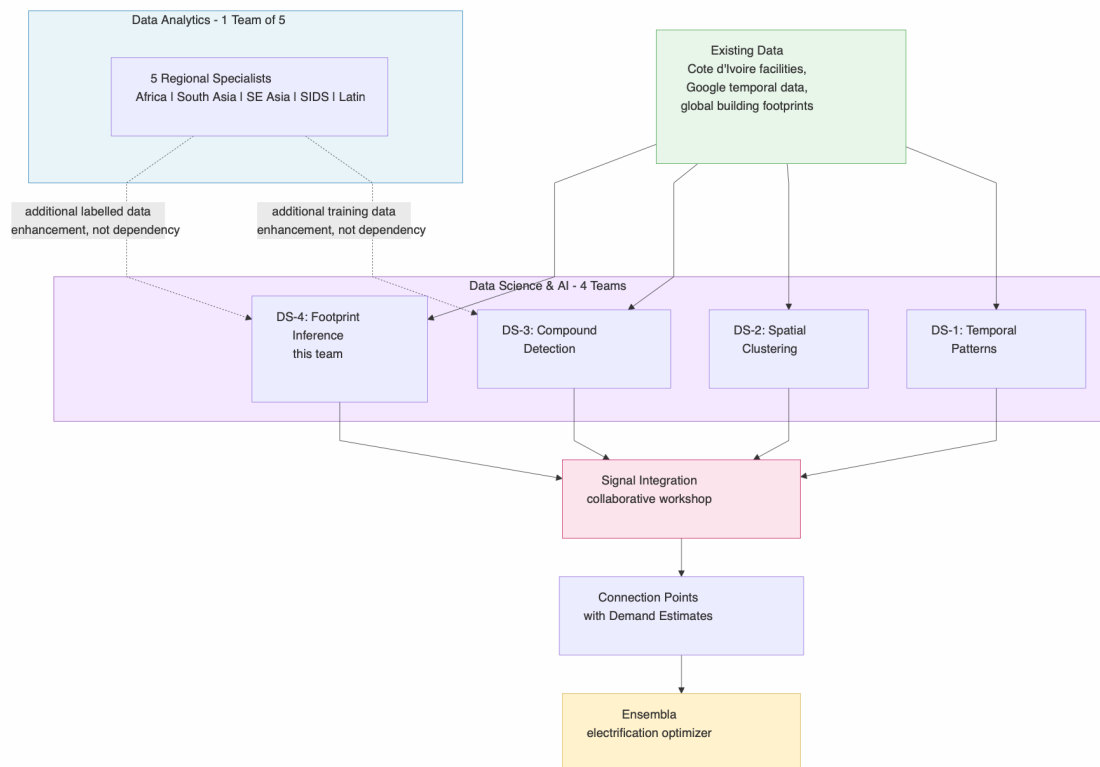
- Which footprint features are most discriminating for building use classification?
- How should class imbalance be handled? (Most buildings are residential; institutional and commercial are rare)
- Do classification models trained in one country transfer to another? (Cote d'Ivoire to Nigeria? To Bangladesh?)
- Which building types are most commonly confused with each other?

This team faces the classic **scale challenge** in applied ML: moving from thousands of labelled examples to classifying 10+ million buildings per country raises real engineering challenges in batch processing, efficiency, and memory management — directly relevant to industry data science roles.

Beyond classification, this team's outputs attach **ESG-enabling metadata** — demand confidence scores, productive use ratios, and temporal growth rates — that feed into downstream financial projection models, connecting building-level ML to institutional investment decisions.

Cross-Team Architecture

This project is one of five complementary proposals submitted to RMIT:



All four DS&AI teams start immediately using existing data. The Data Analytics team discovers additional regional datasets that progressively enrich training data — particularly valuable for DS-4, which gains more labelled buildings from the DA team's facility discovery and DS-3's compound detection work.

From Geospatial Data to Investment Decisions

The RMIT projects sit at the foundation of a chain that connects building-level geospatial data to institutional investment decisions. Here is how the data flows downstream from the work students produce.

Ensembla: Electrification Optimiser

The connection points produced by the combined Data Analytics + Data Science pipeline feed into Ensembla, Prodegee's electrification optimizer. Ensembla determines the optimal electrification strategy for each settlement cluster and costs all viable architectures — solar home systems, DC mesh-grids, mini-grids, or grid extension — presenting the options with their respective costs, sensitivity ranges, and trade-offs rather than prescribing a single "optimal" answer.

Ensembla addresses limitations in the two dominant tools in the field: MIT-Comillas's Reference Electrification Model (REM) and KTH's OnSSET. Neither tool models building-to-

connection mapping (the Fiji Problem that the Data Science teams address), neither evaluates decentralised DC mesh-grids, and both rely on single-metric least-cost optimisation that hides decision-relevant information. The data foundation that the RMIT teams produce provides the building-to-connection mapping these tools lack.

Financial Projection Agent

Ensembla's costed electrification options feed the Financial Projection Agent, which translates project-level technical data into investor-ready financial statements. The agent automates what currently requires manual effort from financial analysts: extracting data from project documents, constructing financial models, and generating reports. It produces profit and loss statements, cash flow forecasts, balance sheets, and key ratios (IRR, NPV, DSCR) — with every assumption logged and auditable.

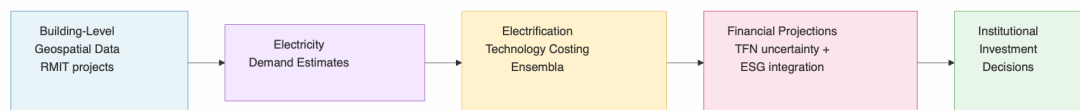
Two methodological innovations distinguish these projections from conventional financial modelling:

Triangular Fuzzy Number forecasting: Instead of presenting point estimates as if they were certain, the methodology uses Triangular Fuzzy Numbers as time-series functions where uncertainty explicitly expands with temporal distance. Climate finance operates in the domain of true uncertainty — known possible outcomes but unknown probabilities — where standard confidence intervals based on historical variance fail. A 20-year cash flow projection honestly shows that year-20 estimates carry far more uncertainty than year-2 estimates, rather than hiding this behind a single discount rate.

ESG-financial synthesis: Rather than treating ESG as a separate reporting exercise bolted onto financial statements, Prodegee's methodology uses ontological relationships to mathematically modulate financial forecasts based on ESG factors. Better environmental performance doesn't just earn a label — it measurably tightens the confidence envelope of financial projections, reflecting the real-world observation that well-governed, environmentally responsible projects have more predictable financial outcomes. ESG factors are integrated through a four-polarities framework that maps sustainability performance to forecast volatility and confidence regions.

These methodologies build on research Prodegee conducted at the Australian National University with funding from the Australian Department of Foreign Affairs and Trade (DFAT), through the ANU Institute for Climate, Energy & Disaster Solutions (ICEDS).

The Complete Chain



The quality of everything downstream depends on the accuracy of the upstream geospatial data that students produce.

Technical Environment

Technology	Role in Project
Python	Primary language — scikit-learn, PyTorch/TensorFlow, GeoPandas
DuckDB	Analytical queries on millions of building records
Kestra	Workflow orchestration — operationalising ML models in production pipelines
Claude Code for VS Code	AI-assisted development environment
PostGIS / PostgreSQL	Spatial database (Supabase)
Git / Gitea	Version control and collaboration

All computing infrastructure, including GPU access for model training, is provided by Prodegee at no cost.

About the Proposer

Marc Torra is originally from Barcelona and brings 25 years of consulting experience in the economic and financial aspects of rural electrification with renewable energy. He has worked across all five regions this project covers — Africa, South Asia, Southeast Asia, Small Island Developing States, and Latin America — with development finance institutions, commercial banks, and project developers. Having seen first-hand how poor data quality undermines investment decisions, he founded Prodegee to apply AI and geospatial analysis to this challenge at scale. He is an RMIT alumnus (Master of International Development), based in regional Victoria, and will serve as the industry supervisor for all five RMIT projects.
